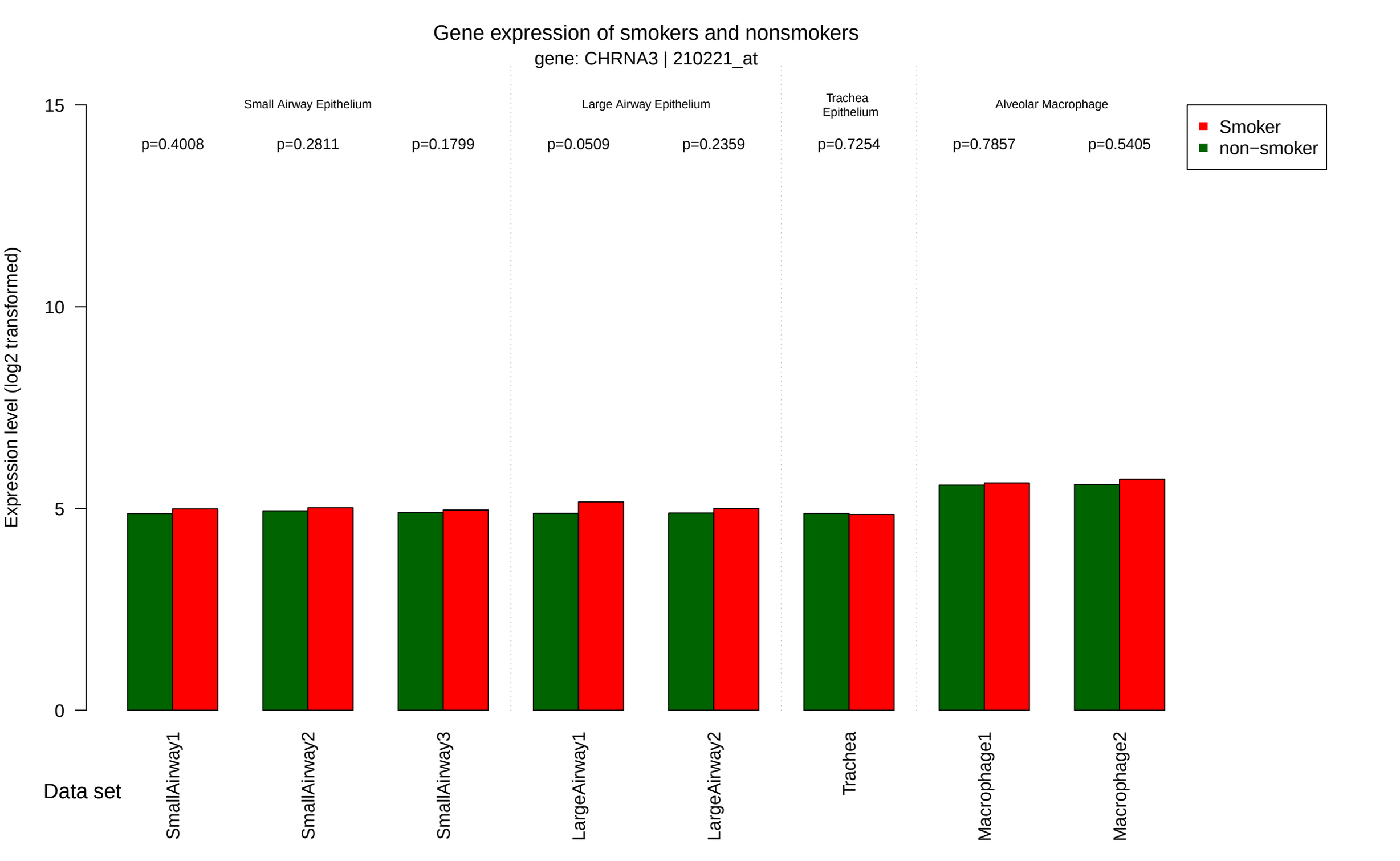
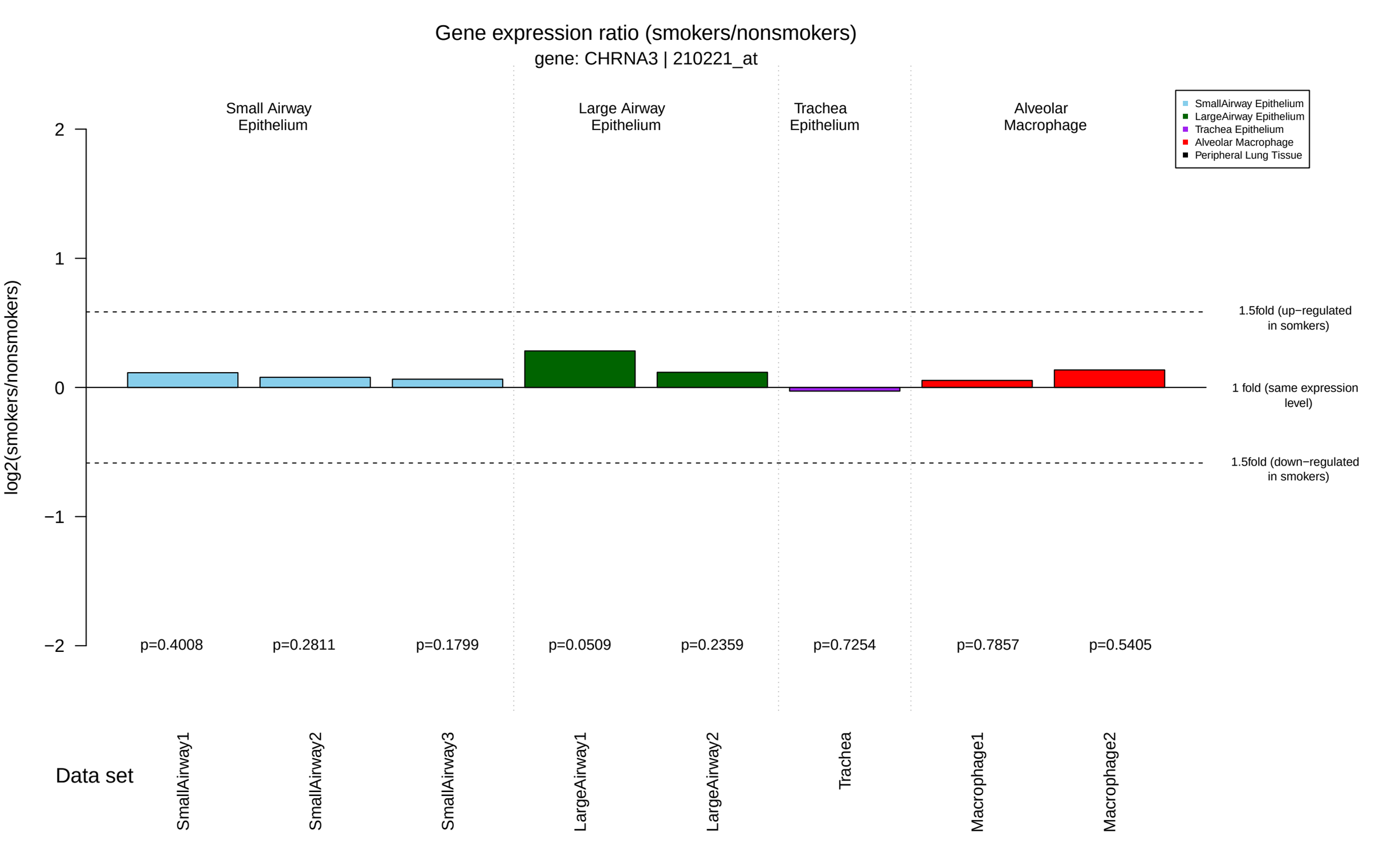
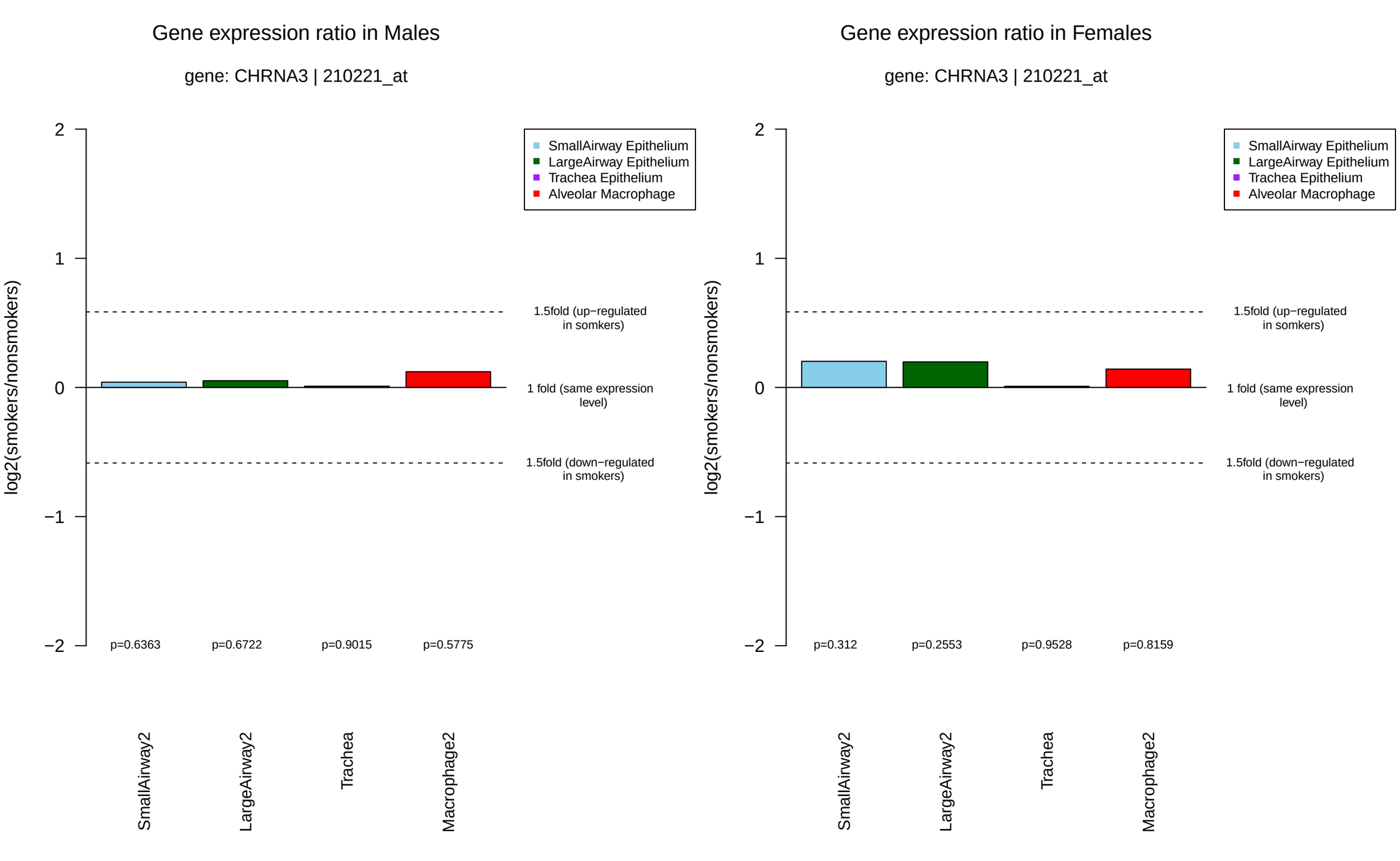
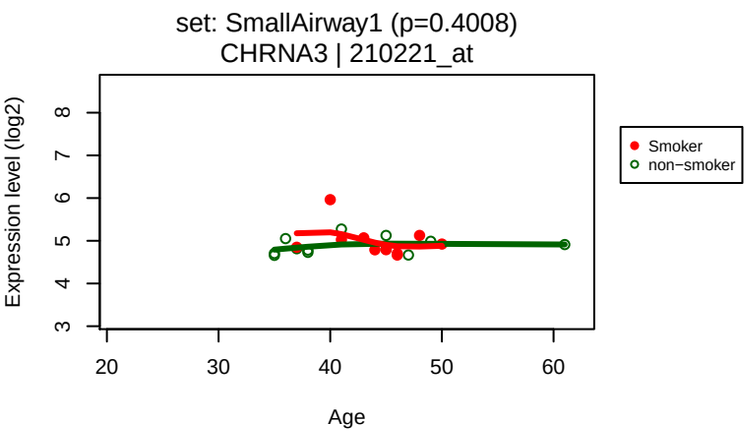
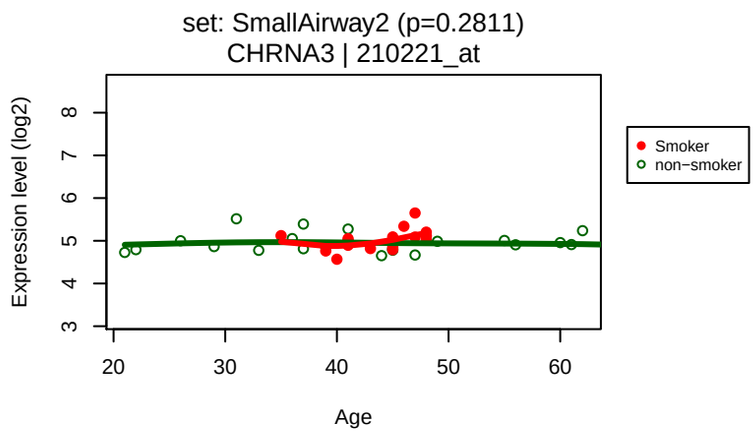
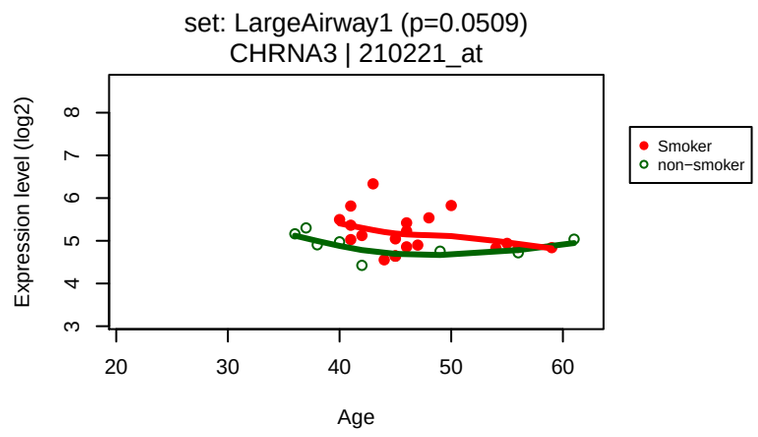

Gene expression of smokers and nonsmokers
gene: CHRNA3 | 210221_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4008
p=0.2811
p=0.1799
p=0.0509
p=0.2359
p=0.7254
p=0.7857
p=0.5405
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

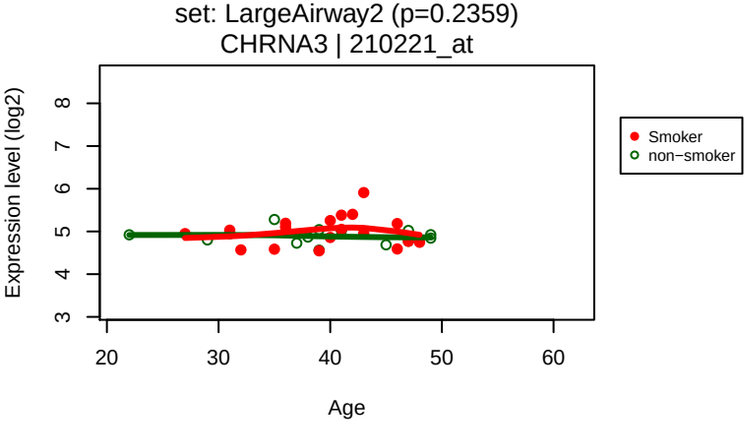

Gene expression ratio (smokers/nonsmokers)
gene: CHRNA3 | 210221_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4008
p=0.2811
p=0.1799
p=0.0509
p=0.2359
p=0.7254
p=0.7857
p=0.5405
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

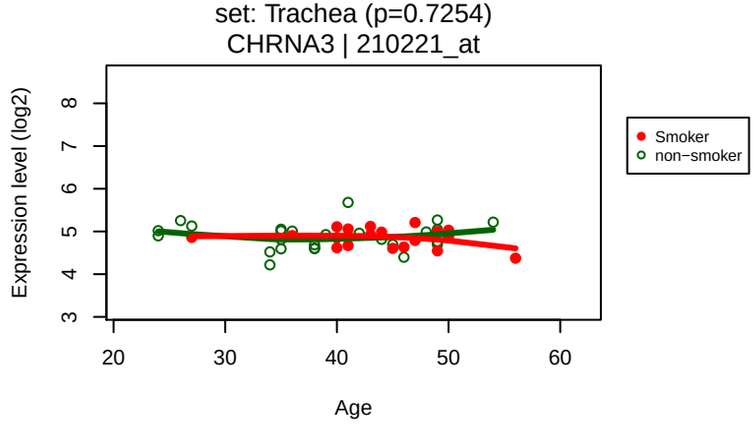

Gene expression ratio in Males
gene: CHRNA3 | 210221_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6363
p=0.6722
p=0.9015
p=0.5775
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHRNA3 | 210221_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.312
p=0.2553
p=0.9528
p=0.8159
SmallAirway2
LargeAirway2
Trachea
Macrophage2

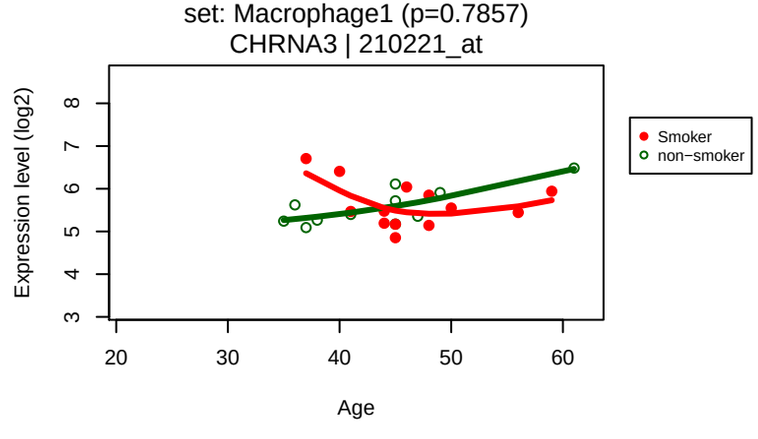

set: SmallAirway1 (p=0.4008)
CHRNA3 | 210221_at
Expression level (log2)
Age
Smoker
non−smoker

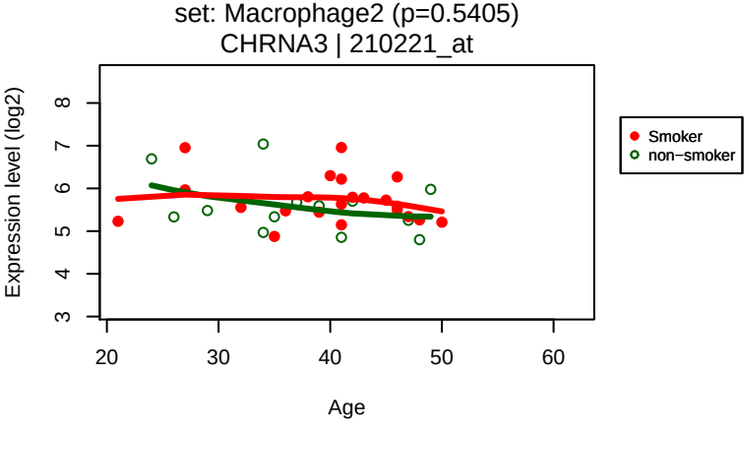

set: SmallAirway2 (p=0.2811)
CHRNA3 | 210221_at
Expression level (log2)
Age
Smoker
non−smoker

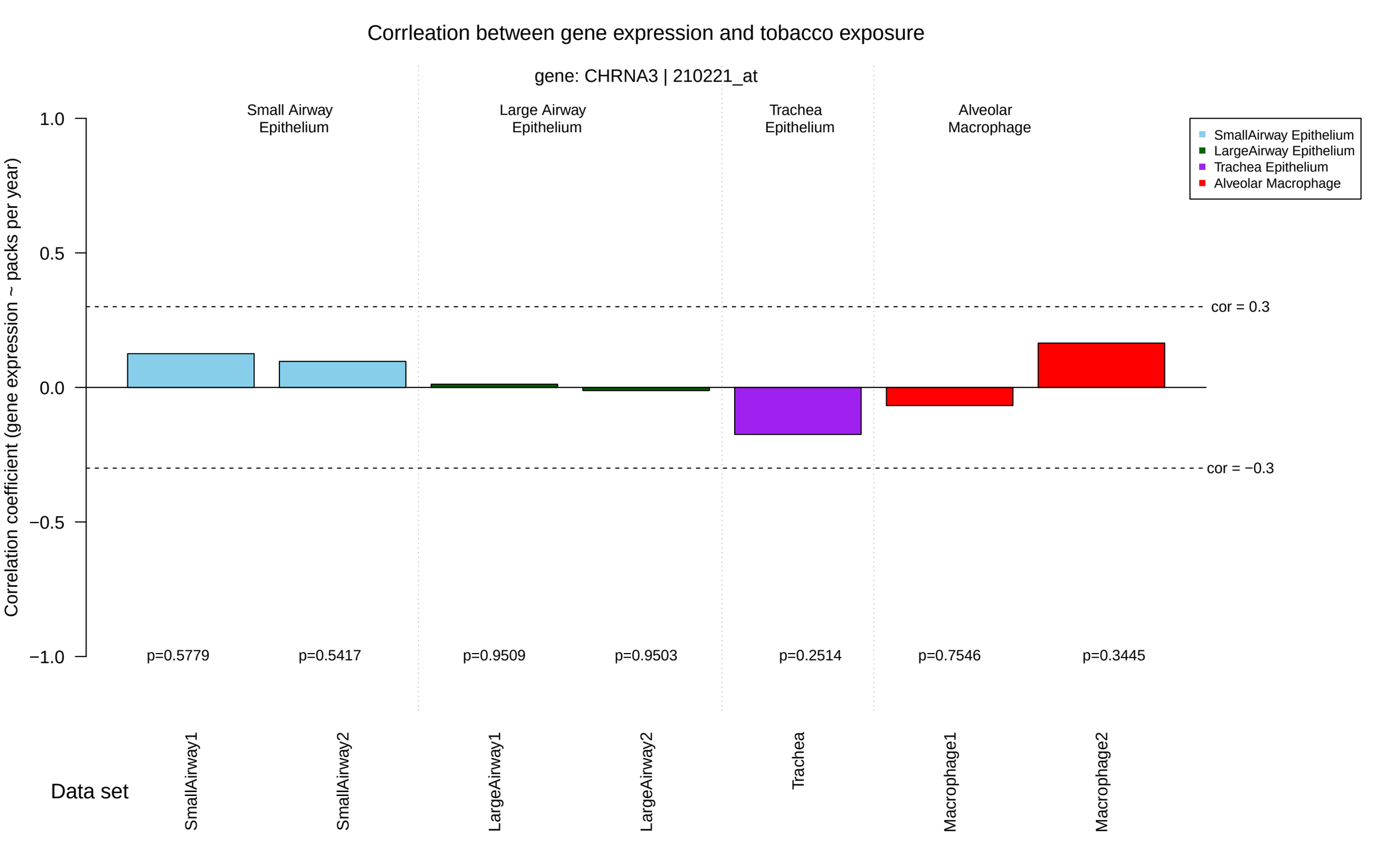

set: LargeAirway1 (p=0.0509)
CHRNA3 | 210221_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2359)
CHRNA3 | 210221_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7254)
CHRNA3 | 210221_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7857)
CHRNA3 | 210221_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5405)
CHRNA3 | 210221_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHRNA3 | 210221_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.5779
p=0.5417
p=0.9509
p=0.9503
p=0.2514
p=0.7546
p=0.3445
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2